

Summary of *What Capable Agents Must Know* (Section 5)

A. Nayebi, arXiv:2603.02491 (2026)

This paper generalises the two Richens world-model papers in three directions it states explicitly: average-case rather than worst-case regret, stochastic rather than deterministic policies, and partial observability, the last addressing an open question raised in *General Agents Contain World Models*. Section 4 shows that the fully observed, Richens-style recovery of the transition kernel survives the first two relaxations. Section 5, summarised here, is the main conceptual step: in a POMDP the agent never sees the latent state, so one cannot in general recover a physical transition kernel from behaviour. What *can* be recovered is the predictive information carried by the history, and the organising principle is

low regret on prediction-sensitive tasks $\implies$ memory preserves task-relevant predictive distinctions.

POMDPs, tests, and predictive state

A finite POMDP is $E = (\mathcal{X}, \mathcal{A}, \mathcal{O}, T, Z, \mu_0)$ with latent state X_t , action A_t , observation O_t , transition kernel $T(x' \mid x, a)$, observation kernel $Z(o \mid x)$ and initial law μ_0 . The agent sees only the history $h_t = (o_0, a_0, o_1, \dots, a_{t-1}, o_t)$, never X_t , so two histories can end in the same observation yet warrant different predictions.

A *test* is a pair $T = (\alpha, W)$: a prescribed future action sequence $\alpha \in \mathcal{A}^k$ and an event $W \subseteq \mathcal{O}^k$ in the resulting observations. Its success probability after h , and its *margin*, are

$$p_T(h) = \mathbb{P}(O_{t+1:t+k} \in W \mid h, A_{t:t+k-1} = \alpha), \quad m_T(h) = |p_T(h) - \tfrac{1}{2}|.$$

A test asks: if these actions are executed, how likely is an observation sequence in W ? For a family $\mathcal{T}$ of tests the *predictive state* is $\eta_{\mathcal{T}}(h) = (p_T(h))_{T \in \mathcal{T}}$. For sufficiently rich $\mathcal{T}$ it is decision-sufficient: the history can be replaced by $\eta_{\mathcal{T}}(h)$ without losing anything needed for optimal decisions. The belief state $b_h(x) = \mathbb{P}(X_t = x \mid h)$ is another decision-sufficient representation, since every $p_T(h)$ is computable from b_h ; a predictive state representation (PSR) instead encodes state through predictions of observable futures rather than a posterior over latent states.

The agent's memory is modelled abstractly as a representation $M = f(h)$ through which the policy factors, $\pi(\cdot \mid h, g_T) = \pi(\cdot \mid M(h), g_T)$; such a policy is *M-based*. The question becomes: which histories may M identify without incurring regret?

Betting goals, margins, and average regret

Prediction is separated from control by an auxiliary *report bit* $B_t \in \{L, R\}$: the agent outputs (B_t, A_t) , the environment ignores B_t , and the bit only exposes the agent's prediction. A test T induces a *betting goal* g_T : L bets that W occurs under α , R that it does not. Writing $q_T(h) = \pi(L \mid h, g_T)$,

$$V^\pi(h; g_T) = q_T(h)p_T(h) + (1 - q_T(h))(1 - p_T(h)), \quad V^*(h; g_T) = \max\{p_T(h), 1 - p_T(h)\} = \tfrac{1}{2} + m_T(h),$$

and the *normalised regret* is $\delta_T(\pi; h) = 1 - V^\pi/V^* \in [0, 1]$. Let the *wrong-action mass* $w_T(h)$ be the probability the policy assigns to the inferior report ($1 - q_T$ if $p_T \geq \frac{1}{2}$, else q_T). The betting case of the paper's Lemma 1 ($\delta = w|u_L - u_R|/\max\{u_L, u_R\}$ with $u_R = 1 - u_L$) gives

$$\delta_T(\pi; h) = w_T(h) \frac{4m_T(h)}{1 + 2m_T(h)},$$

$$\text{hence on tests with } m_T(h) \geq \gamma : \quad w_T(h) \leq \frac{\delta_T(\pi; h)}{c(\gamma)}, \quad c(\gamma) = \frac{4\gamma}{1 + 2\gamma}.$$

This is the basic selection inequality: when one answer is clearly better, low regret forces a stochastic policy to put little mass on the wrong one.

Competence is assumed only on average, $\mathbb{E}_{h \sim \mathcal{H}, T \sim D} \delta_T(\pi; h) \leq \bar{\delta}$. With $E_\gamma = \{(h, T) : m_T(h) \geq \gamma\}$ and $q_\gamma = \mathbb{P}(E_\gamma)$:

Theorem 2 (predictive modelling necessity). For $\gamma \in (0, \frac{1}{2})$,

$$\mathbb{E}[w_T(h) \mathbf{1}_{E_\gamma}] \leq \frac{\bar{\delta}}{c(\gamma)}, \quad \text{equivalently} \quad \mathbb{E}[w_T(h) \mid E_\gamma] \leq \frac{\bar{\delta}}{q_\gamma c(\gamma)} \quad (q_\gamma > 0).$$

So q_γ measures whether the evaluation contains enough informative high-margin tests for average regret to bite. The paper's *non-degenerate evaluation* condition demands mass at least η' on *both* $p_T(h) \geq \frac{1}{2} + \gamma$ and $p_T(h) \leq \frac{1}{2} - \gamma$ (so $q_\gamma \geq 2\eta'$), ruling out near-coin-flip evaluations and constant strategies such as “always report L ”.

Threshold bets and predictive-state recovery

Fair bets reveal only the sign of $p_T(h) - \frac{1}{2}$. **Proposition 1** makes the non-identifiability exact: there are two finite POMDPs (with $|\mathcal{X}| = 4$) on which optimal fair-bet behaviour coincides at every history yet $p_T(h)$ differs for some test, so even exact optimal access to fair bets cannot identify $\eta_{\mathcal{T}}(h)$. One cannot tell $p_T(h) = 0.6$ from 0.99 by watching the decision alone.

To recover magnitudes, introduce a threshold $\lambda \in [0, 1]$. The *threshold betting goal* $g_{T,\lambda}$ offers L , the real test with success probability $p_T(h)$, against R , an independent lottery with success probability λ ; so $V^\pi = q_{T,\lambda} p_T + (1 - q_{T,\lambda})\lambda$, $V^* = \max\{p_T, \lambda\}$, and the optimal report switches at $\lambda = p_T(h)$. On the grid $\lambda_k = (k - \frac{1}{2})/K$, $k = 1, \dots, K$, define

$$\hat{p}_T(h) = \frac{1}{K} \sum_{k=1}^K q_{T,\lambda_k}(h), \quad \epsilon_K = 2\bar{\delta}_K + \frac{1}{4K^2},$$

which for an optimal deterministic agent is the fraction of thresholds lying below $p_T(h)$.

Theorem 3 (predictive-state recovery). If D is supported on tests with $|\alpha| \leq \ell$ and the grid-averaged threshold regret satisfies $\mathbb{E}_{h,T} [\frac{1}{K} \sum_k \delta_{T,\lambda_k}(\pi; h)] \leq \bar{\delta}_K$, then

$$\mathbb{E}_{h,T} [(\hat{p}_T(h) - p_T(h))^2] \leq \epsilon_K,$$

$$\text{and for a finite family } \mathcal{T}_\ell = \{T_1, \dots, T_d\} : \quad \mathbb{E}_h \left[\frac{1}{d} \|\hat{\eta}_{\mathcal{T}_\ell}(h) - \eta_{\mathcal{T}_\ell}(h)\|_2^2 \right] \leq \epsilon_K.$$

The first term of ϵ_K is behavioural error, the second finite-threshold resolution. Sufficiently rich threshold bets therefore recover an approximate predictive-state *vector*, not merely the sign of each prediction.

Theorem 4: recovering predictive dynamics

Theorem 3 recovers the coordinates $p_T(h)$ *at a history*, not a rule for how they evolve. Theorem 4 adds a finite-dimensional linear-PSR assumption. Take core tests $\mathcal{T} = \{T_1, \dots, T_d\}$ and $s(h) = (p_{T_1}(h), \dots, p_{T_d}(h))^\top$. For an action–observation pair $\sigma = (a, o)$ define the shifted test $\sigma \circ T = ((a, \alpha), \{o\} \times W)$ and $s_\sigma(h) = (p_{\sigma \circ T_1}(h), \dots, p_{\sigma \circ T_d}(h))^\top$, and assume linear dynamics $s_\sigma(h) = B_\sigma s(h)$ for all h . Choosing histories $h^1, \dots, h^d$ with $S = [s(h^1) \cdots s(h^d)]$ invertible and $Y_\sigma = [s_\sigma(h^1) \cdots s_\sigma(h^d)]$,

$$Y_\sigma = B_\sigma S, \quad \boxed{B_\sigma = Y_\sigma S^{-1}.$$

Threshold bets estimate S and Y_σ . **Theorem 4** assumes the Theorem 3 regret bound for every test in $\mathcal{T} \cup \{\sigma \circ T_i\}$ and is quantitative: $\|\hat{S} - S\|_F^2 + \sum_\sigma \|\hat{Y}_\sigma - Y_\sigma\|_F^2 \leq d^2(1 + |\mathcal{A}||\mathcal{O}|)\epsilon_K$, and under the conditioning requirement $d\sqrt{(1 + |\mathcal{A}||\mathcal{O}|)\epsilon_K} \leq 1/(2\|S^{-1}\|_2)$,

$$\sum_\sigma \|\hat{B}_\sigma - B_\sigma\|_F^2 \leq C(S, Y) \epsilon_K, \quad C(S, Y) = 8d^2(1 + |\mathcal{A}||\mathcal{O}|) \left(\|S^{-1}\|_2^2 + \|S^{-1}\|_2^4 \sum_\sigma \|Y_\sigma\|_2^2 \right).$$

So Theorem 3 recovers *what the agent predicts now*, while Theorem 4 recovers, under linearity and conditioning, the operators governing how predictive state changes under action–observation pairs, with error vanishing as $\epsilon_K \rightarrow 0$.

Memory necessity: predictive distinctions cannot be aliased

Let $\mathcal{P}$ be a distribution over pairs of histories (h, h') sharing the same last observation, and suppose the memory aliases them, $M(h) = M(h')$. Because the policy factors through M , it must use the same report distribution at both. If a *witness* test $T \in S_\gamma(h, h')$ has $p_T(h) \geq \frac{1}{2} + \gamma$ and $p_T(h') \leq \frac{1}{2} - \gamma$, the confidently correct reports are opposite, so no single M -based policy performs well on both. Define the witnessed aliasing mass $q_\gamma^{\text{Alias}}(M) = \mathbb{P}(M(h) = M(h'), T \in S_\gamma(h, h'))$ and the pair-regret $\bar{\delta}_{\mathcal{P}}(\pi) = \mathbb{E}_{(h, h') \sim \mathcal{P}} \left[\frac{1}{2} (\mathbb{E}_T \delta_T(\pi; h) + \mathbb{E}_T \delta_T(\pi; h')) \right]$.

Theorem 5 (memory necessity). Any M -based policy satisfies

$$\bar{\delta}_{\mathcal{P}}(\pi) \geq q_\gamma^{\text{Alias}}(M) \frac{c(\gamma)}{2}$$

so low pair-regret forces $q_\gamma^{\text{Alias}}(M) \rightarrow 0$: the memory partition must refine the task-relevant predictive-state partition, and histories requiring different confident predictions cannot be collapsed to one internal state. This is the core selection statement. It does not require the agent to store an explicit belief vector or PSR coordinates, only that its internal representation preserve the predictive distinctions the task distribution probes.

Where this sits in the paper

Section 4’s Theorem 1 recovers the fully observed transition kernel under stochastic policies and average regret, $\mathbb{E} |\hat{P}_{ss'}(a) - P_{ss'}(a)| \leq t_\gamma / \sqrt{n} + \bar{\delta} / c(\gamma) + O(1/n)$; its corollaries show this yields Pearl’s Level-2 interventional queries but not, generically, Level-3 counterfactuals. Section 5 replaces physical-state recovery by predictive state, threshold recovery and no-aliasing under partial observability. Section 6 applies the same machinery to structured task distributions (Corollaries 3–5): block-structured tests select informational modularity (no aliasing within each block); a latent regime $I \sim \Lambda$ with $T \sim D_I$ selects persistent regime-sensitive memory, $\mathbb{P}[M(h) = M(h'), I(h) \neq I(h'), T \in S_\gamma] \leq 2\bar{\delta}_{\mathcal{P}} / c(\gamma)$; and under γ -minimality, vanishing-regret agents induce the same decision-relevant partition up to invertible recoding, $M_1 = \varphi(M_2)$. The unifying message is

the evaluation family selects the internal distinctions an agent must preserve.

Two limits for discussion: necessity is task-relative, since different diagnostics select different structure; and the recovery results lean on richness of the test family and, for dynamics, on a linear-PSR structure that general POMDPs need not have.

Source. A. Nayebi, *What Capable Agents Must Know: Selection Theorems for Robust Decision-Making under Uncertainty*, arXiv:2603.02491v3, 2026; Sections 4–6, Lemma 1, Theorems 1–5, Proposition 1, Corollaries 1–5.